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Space Propulsion Project

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Abstract

This study's objective is to examine the heat-transfer environment in the vicinity of the nozzle throat of a Solid Rocket Motor (SRM) by modeling steady-state thermal flux through an actively cooled jacket integrated around the nozzle region.

To achieve the desired performance, a BALLISTIC Test and Evaluation System (BATES)-type motor is designed to deliver 100 kN thrust, 70 bar chamber pressure, and a total impulse of 2.5 MNs, with emphasis on the integrated design of the grain configuration and nozzle geometry. A preliminary sizing of the Thermal Protection System (TPS) is performed for both the combustion chamber and the nozzle sections not protected by the cooling jacket, providing baseline insulation thicknesses and material constraints.

The thermal behavior in the interest region is assessed by tracking the coolant temperature increase through the jacket, which is integrated with a Thermal Barrier Coating (TBC) to mitigate the severe thermal loads concentrated near the throat. Water is selected as the working fluid, entering at 18 °C, and the corresponding coolant mass flow rate and supply pressure are determined to prevent the onset of boiling and maintain fully single-phase operation.

Ultimately, the cooling supply system is designed using a pressure-fed configuration, eliminating the need for external pumps and ensuring independence from auxiliary power sources.

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List of Acronyms

AP Ammonium Perchlorate.

BATES BALListic Test and Evaluation System.

EPDM Ethylene Propylene Diene Monomer.

HTPB Hydroxyl-Terminated PolyButadiene.

MEOP Maximum Expected Operating Pressure.

SRM Solid Rocket Motor.

TBC Thermal Barrier Coating.

TPS Thermal Protection System.

1 Introduction

The development of high-performance SRMs is strongly governed by the management of extreme thermal loads, particularly in the nozzle throat region where heat fluxes are most severe. Conventional TPS, such as ablative materials, are commonly used to protect internal structures; however, they introduce geometry changes during operation, which can affect ballistic performance and limit their suitability for long-duration or reusable ground-test applications.

To overcome these limitations, a BATES-type solid rocket motor is first designed to meet the target performance requirements, providing the baseline configuration for the thermal analysis of the system. The study then investigates an active cooling approach based on a water-cooled jacket integrated around the nozzle region. In addition, a TBC is implemented in combination with the active cooling system to further reduce heat transfer to the structural wall.

2 Literature review

2.1 Thermal Barrier Coating

2.1.1 Thermal environment and heat-flux constraints

Many Rocket propulsion environments impose extreme thermal loads, with hot-gas temperatures T_g regularly higher than 3200 °C and heat fluxes q'' up to more than 100 MW/m² on thrust-chamber walls, motivating rocket-specific TBC designs rather than turbine-derived solutions [1, 2]. In high-pressure thrust chambers, uncoated copper walls experience heat fluxes $q'' \approx 50\text{--}165$ MW/m², with peak values at the throat [3].

The nozzle throat is therefore the most critical region, combining peak heat flux and strong cooling, which generates large thermal gradients. Rocket-relevant studies show that TBCs must withstand $\Delta T > 500$ K and heating rates above 100 K/s, while the substrate is maintained below $\sim 600^\circ\text{C}$ [4].

2.1.2 TBC architecture and material properties

A typical rocket TBC consists of a metallic substrate, a bond coat, and a ceramic top coat. YSZ ($\text{ZrO}_2\text{--}8\text{Y}_2\text{O}_3$) is the most widely used ceramic due to its low thermal conductivity k and strain tolerance [3].

$$k(T) = 2.2 \times 10^{-4}T + 1.09 \quad [\text{W/m} \cdot \text{K}] \quad (1)$$

giving:

$$k_{YSZ} \approx 1.18\text{--}1.57 \text{ W/m} \cdot \text{K}$$

with a typical service limit:

$$T_{\text{service}} \approx 1650 \text{ }^\circ\text{C}.$$

Its performance strongly depends on microstructure (porosity and microcracking), which is controlled by deposition conditions [5].

Advanced multilayer systems such as RSZ/YSZ have been proposed for higher-temperature operation [4]. In these systems, the RSZ top layer provides improved insulation, while YSZ ensures better mechanical compliance.

The thermophysical properties of both materials can be compared as follows.

$$k_{YSZ} \approx 0.90 \text{ W/m} \cdot \text{K}, \quad k_{RSZ} \approx 0.60 \text{ W/m} \cdot \text{K}$$

$$K_{IC,YSZ} \approx 2.2 \text{ MPa}\sqrt{\text{m}}, \quad K_{IC,RSZ} \approx 1.5 \text{ MPa}\sqrt{\text{m}}$$

$$\alpha_{YSZ} \approx 10.7 \times 10^{-6} \text{ K}^{-1}, \quad \alpha_{RSZ} \approx 10.1 \times 10^{-6} \text{ K}^{-1}$$

where K_{IC} , k , and α represent fracture toughness, thermal conductivity, and coefficient of thermal expansion, respectively.

These values indicate that RSZ provides lower thermal conductivity, while YSZ offers higher fracture toughness and slightly higher thermal expansion, contributing to improved thermomechanical compatibility in multilayer architectures.

In practice, TBCs are applied mainly in cooled regions, while uncooled sections rely on ablative or erosion-resistant materials.

2.1.3 Thickness limitations and heat-transfer implications

Heat transfer across the coating follows Fourier's law ¹, which can be expressed in 1D steady-state form as follows:

$$\Delta T = \frac{q'' t}{k} \quad (2)$$

Thin coatings reduce stress but allow higher substrate temperatures, while thick coatings increase stress and spallation risk.

For copper-based systems, $t < 20 \text{ } \mu\text{m}$ is often required at high heat flux [3]. However, multilayer systems on superalloys show different optima. RSZ/YSZ studies report:

$$t_{\text{tot}} \approx 350 \text{ } \mu\text{m}, \quad t_{\text{YSZ}} \approx t_{\text{RSZ}} \approx 175 \text{ } \mu\text{m}$$

achieving 15 cycles at 1800°C [4].

This highlights the trade-off between insulation and mechanical stability.

2.1.4 Thermomechanical failure mechanisms

Failure is dominated by thermal mismatch:

$$\sigma \sim E(\alpha_{\text{tbc}} - \alpha_{\text{metal}})\Delta T \quad (3)$$

leading to delamination and spallation ² [1, 6].

Multilayer studies show:

- thin coatings lead to substrate overheating
- thick coatings lead high energy release rate and spallation

Optimized systems tend toward localized damage rather than catastrophic failure [4].

2.1.5 Nozzle erosion and recession modeling

For uncooled regions, erosion dominates over insulation [7, 8] and the overall nozzle erosion rate is given by the sum of the individual erosion contributions (e.g., chemical, mechanical, etc.):

$$r_{\text{nozzle}}(x) = \frac{1}{\rho_{\text{metal}}} \sum_i r_{i,\text{erosion}} \quad r_c(x) \propto q''(x) \quad (4)$$

Typical throat recession:

$$r_{\text{throat}} \approx 0.128 \text{ mm/s}$$

This confirms that ceramic TBCs are unsuitable outside cooled regions.

¹with ΔT [K] the temperature difference between the two faces of the coating, q'' [W/m²] the surface heat flux, t [m] the thickness of the coating and k [W/(m · K)] the thermal conductivity of the material.

²with α being the coefficient of thermal expansion and E the Young's modulus of the TBC

2.1.6 Comparison table

System	Role	Key properties	Limits	Thickness & Main constraints
YSZ single-layer	Thermal insulation for cooled metallic walls	$k \sim 1.2\text{--}1.6 \text{ W/m}\cdot\text{K}$; good strain tolerance	Limited above $\sim 1600^\circ\text{C}$	Typically $< 200 \mu\text{m}$; too thick $\rightarrow$ spallation, too thin $\rightarrow$ overheating [3]
RSZ/YSZ multilayer TBC	Enhanced insulation for extreme conditions	$k_{\text{RSZ}} \approx 0.60 \text{ W/m}\cdot\text{K}$, $k_{\text{YSZ}} \approx 0.90 \text{ W/m}\cdot\text{K}$	Demonstrated up to 1800°C	$t_{\text{tot}} \approx 350 \mu\text{m}$; optimized ratio required to balance stress and temperature [4]
Metallic coatings	Adhesion + moderate protection	High k (13–80 $\text{W/m}\cdot\text{K}$)	$T_{\text{max}} \sim 1150^\circ\text{C}$	$t \sim 50\text{--}100 \mu\text{m}$; limited insulation capability [1]
UHTCMC (uncooled regions)	Erosion resistance	High-temperature stability	$T_s > 2800 \text{ K}$	mm-scale; suited for uncooled sections [9]

Table 1: Comparison of rocket-nozzle thermal protection approaches.

2.2 BATES motors

2.2.1 Historical context

In the early 1960s, the US Air Force observed that the diversity of industrial test methods made comparing propellant data from various sources difficult. According to H.W. Gale [10], the initial purpose of their program was then to "develop a reproducible, accurate, ballistic evaluation system for solid propellants of interest to the Air Force". This system would allow for a more precise evaluation and comparison of available solid rocket propellants. The first BATES motor firing was successful and took place in 1961. Over the years, the BATES motor has become the universally accepted standard for performance measurement, adopted by US manufacturers and several others abroad.

More recently, the BATES motor has been central to the activities of the NATO RTO Task Group 016. The group's primary objective was to standardize measurement and analysis methods, thereby facilitating international cooperation and reducing overall development costs [11].

2.2.2 Grain geometry

The BATES motor was initially designed for high-fidelity ballistic characterization through the following features developed in [10]:

- **Optimized Combustion:** A short burn time (2–10s) and a high port-to-throat ratio (≈ 7) minimize heat loss, nozzle erosion, and erosive burning, to get the propellant's true burning law.
- **Mass Accuracy:** The absence of insulation and liners to avoid gas flow contamination.
- **Stable Ballistics:** A neutral thrust profile at 1000 psia (68.9 bar) reduces the need for thrust coefficient and tail-off corrections.
- **Aerodynamics:** A slightly sub-optimal expansion ratio ensures full nozzle flow attachment throughout the firing.
- **Geometry & Scale:** The cylindrical grain design eliminates complex geometric effects, while the motor's mass is sufficient to minimize scale-up errors while remaining cost-effective and easy to handle.

The idea of this geometry is the simultaneous radial and axial burning which makes the burning surface A_b nearly constant throughout the firing [12]. Indeed, as the internal radius and so the internal surface increases, the length of the segment L decreases. A relation between the internal and external diameter

of the grain and its length can then be determined to make the initial and final burning area equal, thus guaranteeing a nearly constant pressure in the combustion chamber.

Unlike case-bonded motor in which the propellant is cast directly into the combustion chamber and adheres to it, H.W. Gale [10] highlights that BATES motors have a cartridge-cast system. It allows the grain to be casted separately into a tube, (which serves as an inhibitor) which is then slid into the motor body. This confirms the modular nature of the BATES system: it is designed to perform successive tests quickly by simply replacing the propellant cartridge, while ensuring that test conditions remain strictly identical between the different firings.

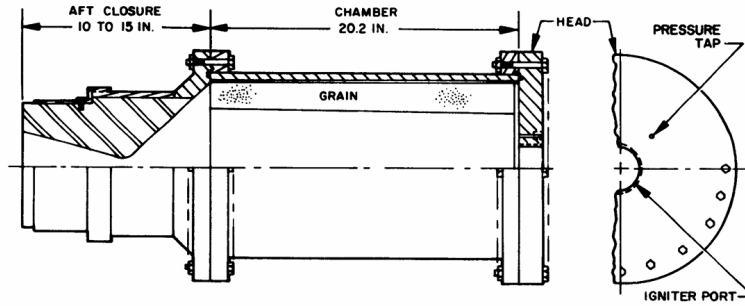


Figure 1: BATES Motor Design Example [10]

2.2.3 Uses of BATES motor

BATES motor technology has been utilized in various applications, some of which are listed below.

Vielle's law: The primary use of the BATES motor is the determination of Vielle's law parameters: $r_b = aP_c^n$, P_c being the combustion chamber pressure and r_b the burning rate.

According to [13], the BATES motor's neutral pressure trace allows for a more accurate calculation of a and n compared to "star-grain" in which the shifting pressure makes it difficult to isolate these constants. Moreover the possibility to use multiple grain geometry makes it more versatile than others complex grain geometries. The ease of fabrication and the reduction of the slivers, unburned propellant residue, are often highlighted as advantages of the BATES geometry compared to star-grain [12].

Two-phases-flow Losses analysis: Real-world performance of aluminized motors is significantly reduced by two-phase flow losses, caused by the low velocity of aluminum oxide particles relative to the gas phase. R. Geisler and C. Beckman [12] explain that BATES motors were used to demonstrate that particle size is primarily driven by the nozzle throat diameter rather than the aluminum content. All the database born from this study was then used to develop industry-standard tools that are still used today: it allows engineers to accurately extrapolate data from small-scale BATES test motors to full-scale boosters.

Radar Attenuation: The BATES motor was used to demonstrate that after-burning in the plume-intensified by aluminum content-creates ionization zones responsible for radar attenuation and telemetry blackouts. At the beginning of the 21st century, this versatility allowed the BATES to be still used for modeling IR/UV signatures and radar cross-section (RCS) for stealth applications [12].

Nozzle erosion: As the BATES motor can maintain a stable pressure over extended durations (up to 25–30 seconds), it has been utilized to test the resistance of nozzle inserts (such as graphite or molybdenum) against thermal and chemical erosion [12].

3 Design of the solid propellant system

According to the design specifications, the SRM must achieve an approximately constant pressure profile and meet the following characteristics:

BATES motor requirements			
Thrust [kN]	T		100
Total Impulse [MN]	T_{tot}		2.5
Chamber Maximum Expected Operating Pressure (MEOP) [bar]	$P_{c,MEOP}$		70

Table 2: Rocket properties

3.1 Propellant composition and performance parameters

The propellant composition is chosen as 78% Ammonium Perchlorate (AP) and 22% Hydroxyl-Terminated PolyButadiene (HTPB), as this oxidizer-to-binder ratio lies within the typical range adopted for composite solid propellants, with bimodal AP consisting of coarse particles (180 μm) and fine particles (100 μm) [14].

The thermodynamical properties of this type of solid propellant, computed using NASA-CEA ³, are listed in the table below:

Combustion properties					
Specific heat ratio [-]	γ	1.25656	Adiabatic Flame T [K]	T_f	2140.3
Molar mass [kg/kmol]	M_m	21.102	Specific heat [kJ/(kg·K)]	c_p	1.89
AP density [kg/m ³] [15]	ρ_{AP}	1950	HTPB density [kg/m ³]	ρ_{HTPB}	920

Table 3: Combustion properties

The propellant overall density ρ_p , considering that 1 kg of propellant is composed of 0.78 kg of AP and 0.22 kg of HTPB, can be computed as:

$$\rho_p = \frac{M_{AP} + M_{HTPB}}{M_{AP}/\rho_{AP} + M_{HTPB}/\rho_{HTPB}} = 1564.6 \text{ kg/m}^3 \quad (5)$$

Once the propellant is fully characterized, it is possible to compute the main performance parameters of the engine. Considering an optimal expansion at sea-level, the exit pressure considered is $P_e = 1$ bar. All the following calculations are performed using a combustion chamber pressure $P_c = 68$ bar. Since the pressure profile of a BATES grain design is not perfectly constant - increasing and then decreasing - a 2 bar margin was applied to ensure the pressure does not exceed the MEOP. This approach is justified in Figure 7, which demonstrates that the maximum pressure reached during the burning process is equal to MEOP.

³Considering frozen chemistry

$$c^* = \frac{\sqrt{\frac{R_u}{M_m} T_f}}{\sqrt{\gamma} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}}} = 1392.9 \text{ m/s} \quad (6)$$

$$c_f = \sqrt{\frac{2\gamma^2}{\gamma-1} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}} \left[1 - \left(\frac{P_e}{P_c} \right)^{\frac{\gamma-1}{\gamma}} \right]} = 1.5680 \quad (7)$$

$$I_s = \frac{c^* \cdot c_f}{g_0} = 222.64 \text{ s} \quad (8)$$

The characteristic velocity reflects the chemical efficiency of the combustion process. Although within the expected range for AP/HTPB in a SRM, the c^* obtained is pretty low for this type of propellant. This is due to the choice of a fuel-rich propellant (as the binder acts as a low-energy fuel) to reduce brittleness and prevent fracturing, as well as the absence of aluminum (high-energy fuel). This choice also leads to a relatively low value of specific impulse (up to more than 240s for aluminized propellants [15]), even though the value stays reasonable.

The mass flow rate and the main geometrical properties of the nozzle can then be computed⁴ :

$$\dot{m} = \frac{T}{I_s g_0} = 45.7862 \text{ kg/s} \quad (9)$$

$$A_t = \frac{\dot{m} c^*}{P_c} = 93.785 \text{ cm}^2 \quad \Rightarrow \quad D_t = \sqrt{\frac{4 \cdot A_t}{\pi}} = 109.28 \text{ mm} \quad (10)$$

$$\varepsilon = \frac{A_e}{A_t} = \frac{1}{\left(\frac{P_e}{P_c} \right)^{1/\gamma}} \sqrt{\frac{2}{\gamma-1} \left[\left(\frac{P_c}{P_e} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right]^{\frac{\gamma+1}{2(\gamma-1)}}} = 7.9643 \quad (11)$$

While a vacuum nozzle typically features a higher expansion ratio, a value of $\varepsilon = 8$ is common for sea-level nozzles to avoid the over-expansion phenomenon, which occurs when the exit pressure falls below atmospheric pressure [15].

3.2 Ballistic properties

In order to estimate Vieille's law coefficient ($r_b = a \cdot P_c^n$), a series of BATES motor firings is performed, from which burning rates are measured at different pressures:

P_c	10.1	9.7	10.3	30.2	31.0	29.8	50.2	51.0	50.3	70.2	69.0	69.8	91.2	89.1	89.0
r_b	4.0	3.8	4.1	5.6	6.0	5.7	7.0	7.2	7.1	8.4	8.3	8.6	8.8	9.0	9.2

Table 4: Chamber Pressure (P) [bar] and burning rate (r_b) [mm/s] measurements

The a and n coefficients can be estimated by linearly interpolating the logarithmically transformed data.

⁴ A_t throat area, A_e exit area, ε expansion ratio

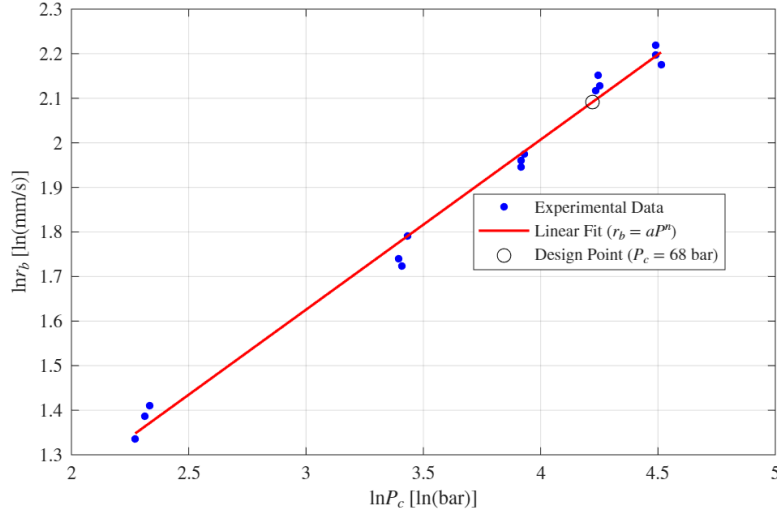


Figure 2: Vieille's law interpolation

$$\log(r_b) = n \log(p) + \log(a) \xrightarrow{\text{linear interpolation}} \begin{cases} a = 1.6177 \frac{\text{mm/s}}{\text{bar}^n} \\ n = 0.3816 \end{cases} \quad (12)$$

The burning rate at the design pressure is then interpolated to get $r_b = 8.0944$ mm/s.

3.3 Combustion chamber sizing

3.3.1 Grain design

The shape chosen for the grain is based on the principle of BATES motors. As described previously, it is a tubular shape that burns both radially and axially, thus reducing the length and increasing the internal diameter of the grain at the same time. The combination of effects that result in a quasi-constant burning area allows the motor to provide a nearly constant thrust during the combustion process.

The burning area of this type of geometry can be computed as:

$$S_b(t) = 2\pi R_i(t) \cdot L(t) + 2\pi(R_e^2 - R_i(t)^2) \quad (13)$$

With $L(t)$, $R_i(t)$, and R_e denoting the grain length, internal radius, and external radius, respectively. Considering a burning rate r_b , $L(t)$ and $R_i(t)$ are calculated as follows:

$$L(t) = L_{init} - 2r_b t \quad \text{and} \quad R_i(t) = R_{i,init} + r_b t \quad (14)$$

Since the burning area evolves quadratically with time, $\frac{dS_b}{dt}$ cannot be null for the whole duration of the burning time, thus this geometry cannot achieve a perfectly constant thrust profile. It initially increases (dominated by the expansion of the internal port) before decreasing (driven by the shortening of the grain).

To achieve global neutrality, the initial and final burning areas must be the same. This condition will result in a relation between the length and the radius of the grain:

$$\begin{aligned}
 S_{b,init} &= S_{b,final} \\
 \Rightarrow 2\pi R_{i,init} L_{init} + 2\pi(R_e^2 - R_{i,init}^2) &= 2\pi R_e L(t_b) \\
 \Rightarrow L_{init} &= 3R_e + R_{i,init}
 \end{aligned} \tag{15}$$

Satisfying this condition ensures that the initial and final surface areas are the same.

By considering the thickness to be null at the end of the burning time, and assuming constant r_b since for a BATES motor the chamber pressure can legitimately be constant in first approximation, the initial web thickness can be computed based on the burning rate at 68 bar and the burning time $t_b = 25$ s:

$$w = r_b(P_c = 68 \text{ bar}) \cdot t_b = 202.3 \text{ mm} \tag{16}$$

The external and internal radii are then linked by the relation $R_e = R_i + w$.

Knowing the mass flow rate $\dot{m}$ required to provide a $100kN$ thrust, the density ρ_p of the propellant and the burning rate, the ideal burning area A_b can be computed.

$$A_b = \frac{\dot{m}}{\rho_b r_b} = 3.6159 \text{ m}^2 \tag{17}$$

By solving $S_b(0) = A_b$ yields the internal initial radius $R_{i,initial}$, from which the external radius R_e and length L_{init} are derived using the aforementioned relations. The grain geometry is therefore summarized in Table 5.

Grain geometry			
Initial internal diameter D_i [mm]	507.4	External diameter D_e [mm]	912.0
Web thickness w [mm]	202.3	Initial length L_i [mm]	1621.8

Table 5: Grain geometry

The validation of this design is based on the calculation of Klemmung, denoted as $K_n = \frac{A_b}{A_t}$, presented in Figure 3. Since the throat area A_t remains constant for a non ablative nozzle, the evolution of K_n directly follows that of the burning surface area A_b . The grain geometry was designed to be neutral, thereby ensuring a nearly constant burning area and K_n throughout the firing, with identical values at ignition and burnout.

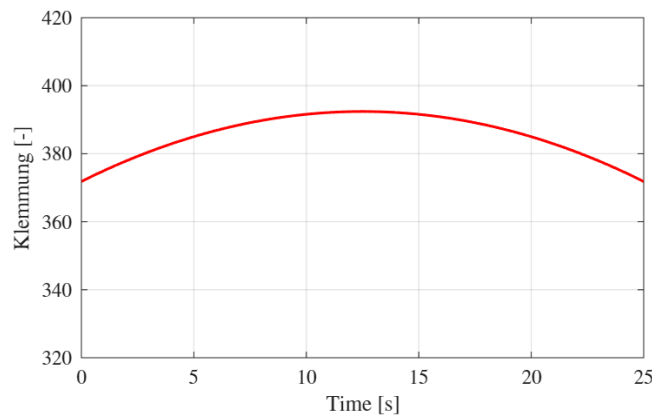


Figure 3: Evolution of Klemmung K_n throughout the firing

The burning rate considered in this part does not take into account erosive burning, which is the phenomenon responsible for the increase of the propellant burning rate in high velocity or high mass flow rates. According to [15], this effect can be neglected for a port-to-throat area⁵ ratio A_p/A_t greater than 4. The design obtained gives an initial value of $A_p/A_t = 21.56$, which will increase throughout the burning process, thus validating the initial hypothesis of considering only the non-erosive burning rate.

3.3.2 Combustion chamber insulation

The combustion chamber must be covered with a thermal protective layer that ablates during the combustion process, for which a typical choice in solid rocket motor cases is Ethylene Propylene Diene Monomer (EPDM).

The regression rates for this material are retrieved from [16], where EPDM is tested under significantly harsher conditions (3281 K with aluminized propellants) than those considered here. Therefore, $r_{\text{EPDM}} = 0.163 \text{ mm/s}$ can be considered a conservative assumption; with the addition of a safety factor of 1.25, it ensures that complete erosion of the EPDM layer is avoided, and the resulting thickness is computed by considering a maximum exposure time of 25 s [17]:

$$t_{\text{EPDM}} = r_{\text{EPDM}} \cdot t_b \cdot SF = 5.09 \text{ mm} \quad (18)$$

The thickness of the EPDM layer was assumed to be constant along the chamber using this worst-case value. This choice is justified by the fact that the system is a test motor not subject to strict mass constraints, and the primary objective is to achieve a constant thrust profile; therefore, the geometry has been simplified as much as possible. Additionally, insulation is typically applied using rolls of constant thickness, making it difficult to achieve axial scaling of the EPDM.

3.3.3 Casing design

The choice of the material used for the casing of the combustion chamber is the result of a trade-off that is made between weight, internal pressure, and simplicity of fabrication. AISI 4340 was selected for its high-strength properties and ease of manufacturing since its high density is not a main concern, as this motor is not intended to fly. As mentioned in [18], this is a well-known alloy that has been used for the past decades in high-strength rocket cases. To get optimal performances and a capacity of absorbing energy without breaking, a longitudinal specimen tempered at 537°C was chosen, with a yield strength $\sigma_y = 1144 \text{ MPa}$ at ambient temperature.

The casing of the combustion chamber must be designed to withstand the internal pressure. As a first estimation, the simplified membrane theory can be used to approximate the thickness of the casing. The two main stresses that have to be considered for the sizing of a cylinder-shaped tank are the axial stresses σ_l and the hoop stresses σ_θ , thanks to the following formula [15]:

$$\sigma_\theta = 2\sigma_l = \frac{P_c R_{i,cas}}{t} \quad (19)$$

t being the thickness, and $R_{i,cas}$ the internal radius of the casing which can be obtained as:

$$R_{i,cas} = R_e + t_{\text{EPDM}} = 461.11 \text{ mm} \quad (20)$$

Taking into account the reduction in σ_y of the AISI 4340 due to exposure to high temperature and the uncertainties on the efficiency of the thermal protection, a safety factor $\eta = 1.5$ is applied. The thickness of the casing is then obtained.

⁵with $A_p = \pi R_i^2$

$$t_{cas} = \frac{P_{c,MEOP} R_{i,cas} \eta}{\sigma_y} = 4.23 \text{ mm} \quad (21)$$

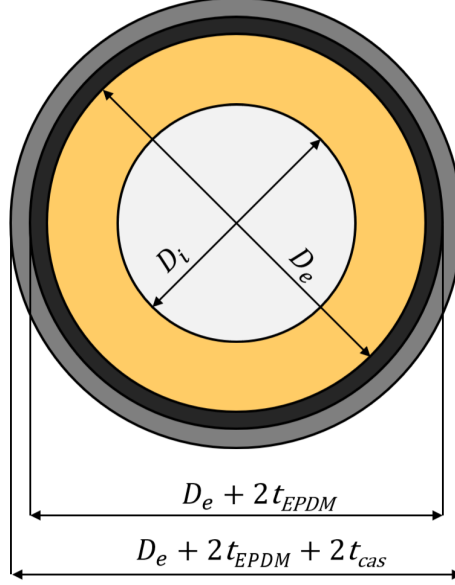


Figure 4: Axial cross-section of the combustion chamber

3.4 Design of the nozzle geometry

To ensure a clear understanding of the flow process and the subsequent design choices, the nozzle can be divided into two main functional regions: the convergent section and the divergent section. This division reflects the transition between the subsonic and supersonic regimes, where geometric parameters influence engine performance in profoundly different ways.

The convergent section serves to accelerate the exhaust gases from the combustion chamber to sonic conditions at the throat. Because this geometry has a negligible impact on the final supersonic expansion efficiency, its design was kept straightforward. A standard conical configuration was selected, defined by the external diameter of the grain (D_e), the throat diameter (D_t), and a convergent half-angle (β) of 45° . This provides an optimal compromise between a smooth subsonic flow approach, structural compactness, and ease of manufacturing. The required length of the convergent section (L_{conv}) is therefore determined by:

$$L_{conv} = \frac{D_e - D_t}{2 \tan(\beta)} = 435.33 \text{ mm} \quad (22)$$

Consequently, geometric optimization focuses entirely on the divergent section, which expands the flow from the throat (D_t) to the final exit diameter (D_e). For this region, a Rao (bell-shaped) contour was selected over a standard conical geometry to turn the exhaust flow closer to the axial direction.

To define the specific parabolic profile, Rao's approximation method was implemented using charts taken from [17]. The shape of the bell is determined by establishing the initial wall angle at the inflection point (θ_i) and the final exit angle (θ_e). Constraining the design to a 100% fractional length—equivalent to the axial length of a standard 15° half-angle conical nozzle—yielded the following geometric parameters and divergent length ($L_{div,Rao}$):

$$L_{div,Rao} = L_{div,conical} = \frac{D_e - D_t}{2 \tan(15^\circ)} = 398.84 \text{ mm} \quad \theta_i = 20.96^\circ \quad \theta_e = 6.77^\circ \quad (23)$$

The design of the throat fillets is critical in this process to ensure a smooth flow transition and prevent boundary layer separation. Specifically, a fillet radius of $1.5R_t$ (throat radius) was selected for the convergent section, and $0.382R_t$ for the divergent section, as given by Rao's approximation method.

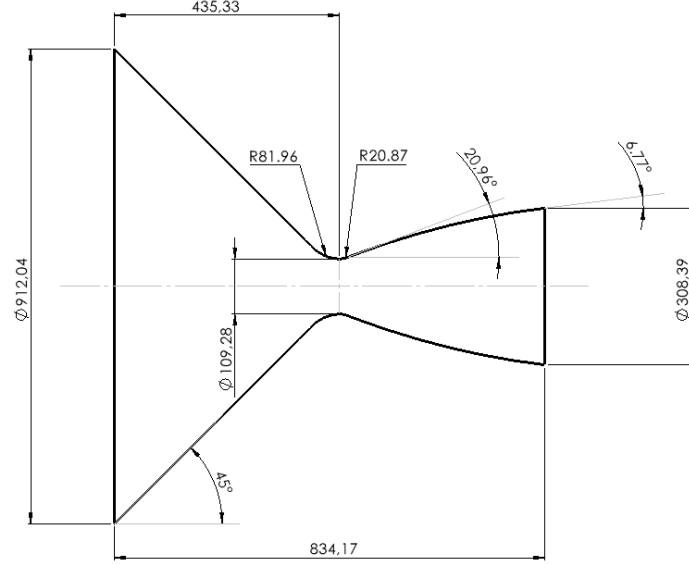


Figure 5: Nozzle internal geometry (dimensions in [mm])

With the geometric parameters established, the divergence loss factor can be evaluated for both the conical and Rao configurations. Since this loss is fundamentally driven by the non-axial components of the exhaust flow, this comparison relies on a 2D thrust evaluation.

$$\lambda_{conical} = \frac{1 + \cos(15^\circ)}{2} \implies T_{2D,conical} = \lambda_{conical} \cdot T_{ideal} = 98296.29 \text{ N} \quad (24)$$

$$\lambda_{Rao} = \frac{1}{2} \left[1 + \cos \left(\frac{15^\circ + \theta_e}{2} \right) \right] \implies T_{2D,Rao} = \lambda_{Rao} \cdot T_{ideal} = 99100.55 \text{ N} \quad (25)$$

Consequently, the adoption of the Rao contour provides a net 2D thrust increment of 804.26 N over the baseline conical configuration by effectively minimizing radial divergence losses.

These characterizations regards only the internal geometry of the nozzle. Considerations on the wall thickness, thermal protection and materials choice will be done after a preliminary thermal analysis in subsection 4.6 and subsection 4.7.

Throat erosion

The phenomenon of throat erosion can lead, in many cases, to the modification of the throat area, thus leading to the modification of the performances during the firing ($P_c \propto \frac{A_b}{A_t}$). This effect is mainly due to the aluminum particles added to the propellant which induce mechanical ablation through the direct bombardment of the nozzle throat, physically eroding the material surface in addition to the thermal and chemical degradation. However, the propellant chosen does not contain any aluminum, the throat erosion was then neglected and the geometry of the nozzle considered as constant throughout the firing.

3.5 Internal flow characterization of the nozzle

To complement the geometric design, the internal flow behavior was analyzed by evaluating the distribution of the main thermodynamic properties along the nozzle axis. This characterization was carried out under the assumptions of one-dimensional (1D) isentropic flow and frozen flow conditions. The isentropic model assumes a steady, adiabatic, and reversible expansion, while the frozen flow assumption implies that the chemical composition of the exhaust gases remains constant, neglecting recombination effects during the expansion.

The local Mach number $M(x)$ at any axial position is intrinsically governed by the nozzle geometry. Under the 1D assumption, it is evaluated using the Area-Mach relation:

$$\frac{A(x)}{A_t} = \frac{1}{M(x)} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M(x)^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}} \quad (26)$$

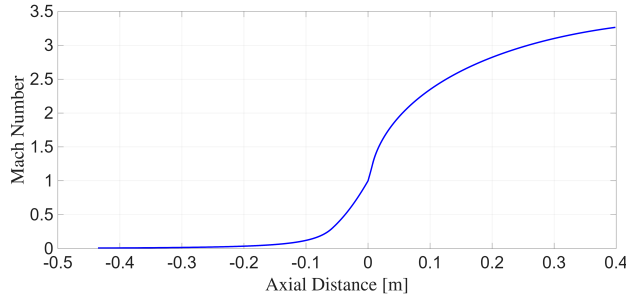
where $A(x)$ is the local cross-sectional area, A_t is the throat area, and γ is the specific heat ratio of the exhaust gas mix. Since this transcendental equation cannot be inverted explicitly for M , the Mach number distribution was solved numerically at each axial station using the **fzero** iterative solver in MATLAB, ensuring a residuals-based convergence for both the subsonic and supersonic regimes.

Once the Mach number distribution is resolved, the static flow properties of the hot gas are obtained utilizing the standard isentropic relations:

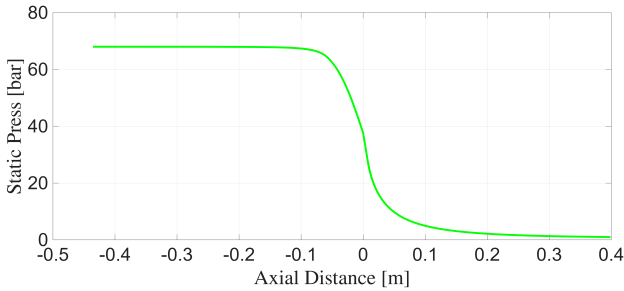
$$T(x) = \frac{T_0}{1 + \frac{\gamma - 1}{2} M(x)^2} \quad (27)$$

$$p(x) = p_0 \left(1 + \frac{\gamma - 1}{2} M(x)^2 \right)^{-\frac{\gamma}{\gamma - 1}} \quad (28)$$

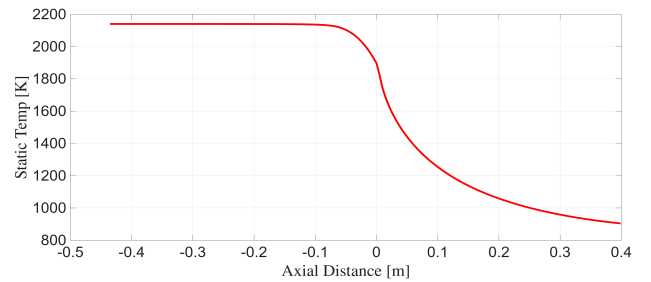
where T_0 and p_0 represent the stagnation temperature and stagnation pressure established inside the combustion chamber. The resulting axial profiles computed through this analytical framework are illustrated in Figure 6.



(a) Mach number distribution



(b) Static Pressure



(c) Static Temperature

Figure 6: Distribution of flow properties along the nozzle axis.

The resulting profiles for the Mach number, static pressure, and static temperature (Fig. 6) demonstrate a smooth and consistent expansion process, where the flow accelerates to sonic conditions ($M = 1$) at the throat and continues to reach supersonic speeds throughout the Rao divergent section. This acceleration, coupled with a continuous decrease in static pressure and temperature, confirms the effective conversion of the gas's internal energy into kinetic energy within the optimized bell-shaped contour.

3.6 Real performance

Once the complete geometry of both the grain and the nozzle has been designed, the thrust and pressure traces for a single firing can be computed by modeling the propellant regression within the case.

Assuming an initial ignition pressure of 30 bar [19], the burn rate r_b can be computed using Vieille's law. By considering a time step of 0.01 s, the propellant grain is regressed by $r_b \cdot t_{step}$ from both ends as well as from inner radius. In this way, the burning area A_b at the next time step can be computed from the updated grain geometry, and the corresponding chamber pressure is then given by ⁶:

$$P_c(t_{step}) = \left(a \rho_p c^* \frac{A_b(t_{step})}{A_t} \right)^{\frac{1}{1-n}} \quad (29)$$

From this P_c , a new burning rate is computed, and the iterative process is repeated until the web thickness reaches zero.

The thrust profile is then directly obtained from the resulting pressure evolution:

$$T(t_{step}) = A_t c_f P_c(t_{step}) \quad (30)$$

Once this thrust trace is available the real one can be computed by simply applying the 2D losses coefficient due to divergence at the exit of the nozzle shown in subsection 3.4.

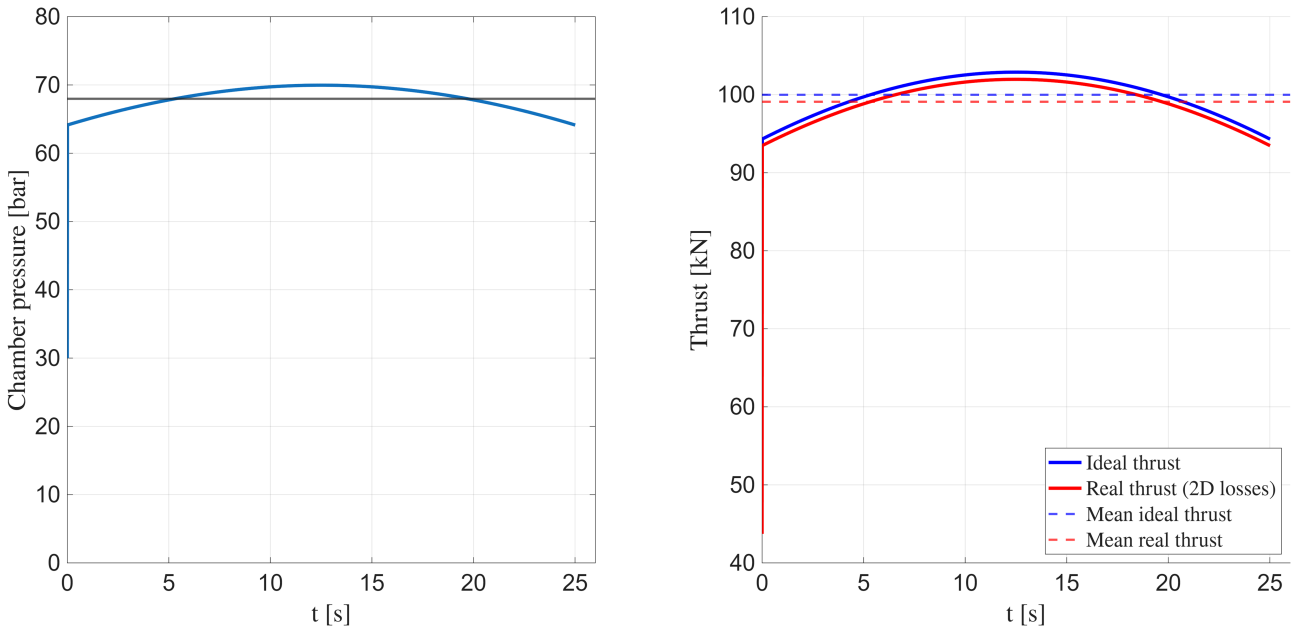


Figure 7: Real pressure and thrust profile

⁶Assuming all parameters to be constant except for A_b

The ideal and real specific impulse can be obtained by computing the mass flow rate at every time step as:

$$\dot{m} = A_b \rho_p r_b \quad (31)$$

I_s shows an almost constant behavior as can be seen in Figure 8 and decreases by ≈ 2 s when losses are introduced.

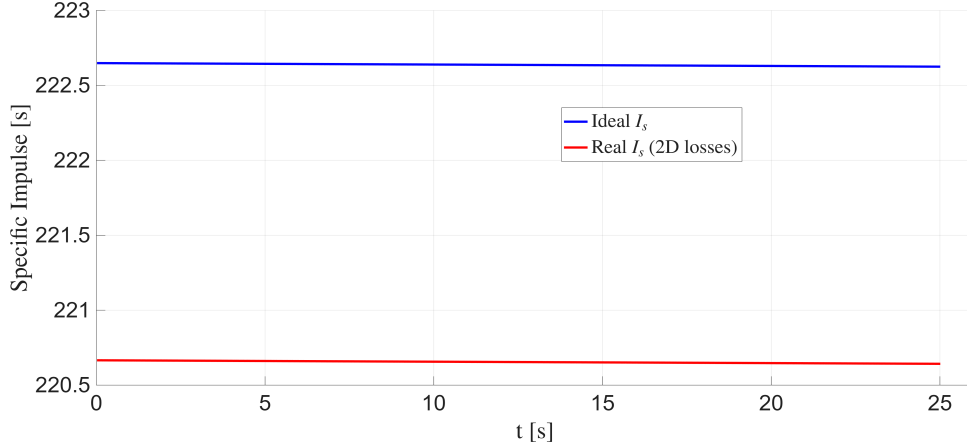


Figure 8: Real specific impulse

The total impulse can be computed in both cases by numerically integrating the respective thrust trace:

$$I_{tot} = \int_0^{t_b} T \, dt \quad \Rightarrow \quad \begin{cases} I_{tot,ideal} = 2.5004 \text{ MNs} \\ I_{tot,real} = 2.4781 \text{ MNs} \end{cases} \quad (32)$$

As shown in Figure 7, this grain and nozzle design meets all the requirements, providing a quasi-constant thrust with a mean value of 100 kN, a MEOP of 70 bar, and a total impulse of 2.5 MNs.

Parameter	Ideal	Real	Relative difference
Max. Thrust [kN]	102.89	101.97	-0.9%
Specific Impulse [s]	222.63	220.65	-0.9%
Total Impulse [MNs]	2.5004	2.4781	-0.9%

Table 6: Ideal and Real performance comparison

The pretty small difference (less than 1%) between ideal and real performance can be explained by the choice of a long bell nozzle (100% of the traditional 15° half-angle conical nozzle) to reduce the divergence losses. This design choice was feasible as the overall dimensions of the motor were not a primary constraint.

3.7 Estimation of uncertainty

The uncertainty on the MEOP and burning time is estimated through a Monte Carlo campaign, in which the regression law coefficients a and n , as well as the web thickness w , are varied within their respective uncertainty ranges. Moreover, this analysis assumes a constant ideal c^* , so that variations in chamber pressure are directly related to uncertainties in the burning rate and grain geometry.

Vieille's law coefficients, being derived from experimental data fitting, inherently include measurement

and interpolation errors and represent the largest source of uncertainty. Since the linear regression used to obtain these coefficients is performed in logarithmic space:

$$\log(r_b) = n \log(p) + \log(a) \quad (33)$$

only n and $q = \log(a)$ can be appropriately described by a Gaussian distribution with:

$$\mu_n = 0.3816 \quad \sigma_n = 0.0106 \quad \mu_q = 0.4810 \quad \sigma_q = 0.0399 \quad (34)$$

the distribution of a can be recovered by inverting the logarithmic transformation:

$$a \sim \exp\left(\mathcal{N}(q, \sigma_q^2)\right) \quad (35)$$

Although, other model parameters are also affected by uncertainties, only the web thickness has been considered in this analysis as it has a first-order effect on the burn duration.

Assuming a 0.5 mm 3- σ manufacturing precision [20]:

$$\mu_b = 202.3 \text{ mm} \quad \sigma_b = 0.1667 \text{ mm} \quad (36)$$

Using the same approach as in subsection 3.6, 10,000 Monte Carlo simulations are performed by sampling random values of these parameters according to their respective distributions, resulting in the following MEOP and t_b distributions:

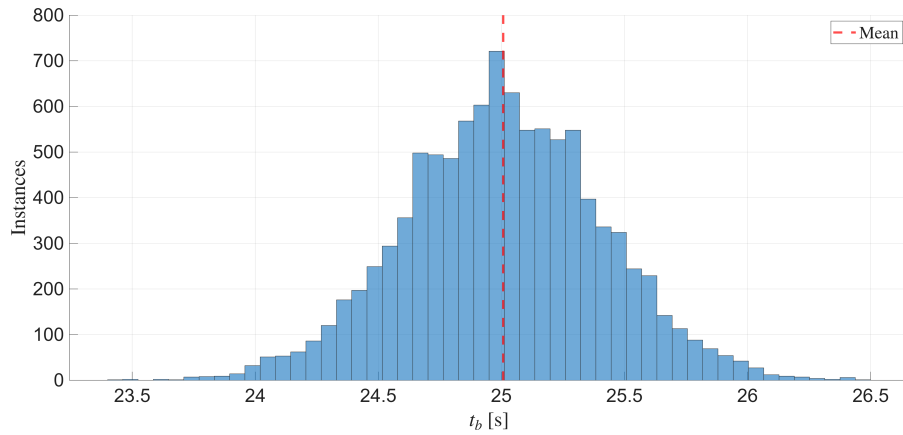


Figure 9: Distribution of t_b

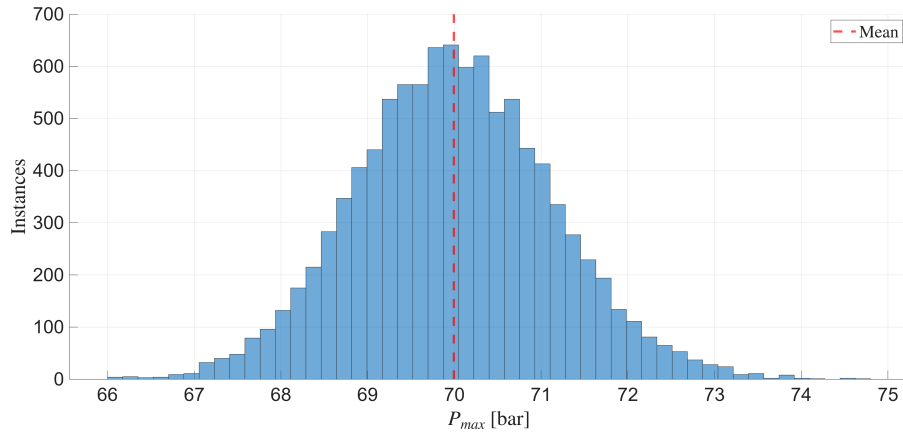


Figure 10: Distribution of P_{Max}

The resulting distributions show that the mean of both MEOP and t_b coincide with their design targets. Additionally it's possible to observe that in all cases the results lie within a $\pm 7\%$ window around the nominal values, demonstrating that even in the presence of uncertainties this architecture obtains performances close to the design point.

4 Thermal analysis and implications

A thermal analysis of the nozzle is carried out to quantify the heat loads and assess their impact on the cooling system design, material limits, and structural integrity.

4.1 Requirements for the design of the cooling jacket

The cooling system must be designed to enable the estimation of the heat flux through measurement of the coolant temperature rise, and to achieve this objective several requirements are imposed.

In particular the cooling jacket operates in the vicinity of the nozzle throat, spanning from an area ratio of 2 in the subsonic region to an area ratio of 2 in the supersonic region.

Furthermore water is selected as the coolant, entering the jacket at a temperature of 18°C , and the system is required to operate for the entire burn duration, including a safety margin of 2 seconds before ignition and after shutdown. Throughout the operation, the coolant must remain in the liquid phase to avoid boiling onset, while the wall temperatures must be maintained within acceptable limits. Finally, both the cooling jacket geometry and the associated flow parameters must be compatible with manufacturing constraints.

4.2 Computational model for the cooling jacket

4.2.1 Goal and assumptions

The goal of the computational model is to estimate the design parameters and performance of the cooling jacket, and to accomplish this, a number of simplifying assumptions are introduced.

First of all, the analysis is conducted under steady-state conditions, thereby neglecting thermal transients, and the flow in the nozzle is treated as one-dimensional, compressible, and isentropic with constant specific heat.

Additionally, heat transfer through the wall is assumed to be purely radial and one-dimensional, neglecting axial conduction, while the TBC and the metal wall are modeled as homogeneous layers with constant thermal conductivity. Furthermore, radiative heat transfer is neglected as a first approximation, and this assumption is justified a posteriori after the computations are completed.

On the coolant side, the flow is considered single-phase, incompressible, and fully developed turbulent along the entire channel; accordingly, both velocity and thermophysical properties are assumed constant, with the latter evaluated at a reference temperature defined as the midpoint between the inlet and saturation temperatures at the operating pressure.

Ultimately, pressure losses in the cooling channel are estimated using a Darcy friction factor correlation for turbulent flow, and the cooling jacket is discretized along the axial direction into finite segments matching the nozzle geometry, over which a one-dimensional energy balance is applied sequentially.

4.2.2 Model overview

The computational model is explained below, assuming the presence of a single layer TBC. The model for a bare wall follows the same logical framework.

The local thermodynamic properties of the hot gas, specifically the static temperature $T(x)$ and static pressure $p(x)$, are derived from the isentropic relations previously defined in subsection 3.5 (Equation 27 and Equation 28). To complete the characterization of the flow field for the heat transfer analysis, the local gas density $\rho(x)$ is determined as follows:

$$\rho(x) = \rho_0 \left(1 + \frac{\gamma-1}{2} M(x)^2 \right)^{-\frac{1}{\gamma-1}} \quad (37)$$

where ρ_0 is the stagnation density in the combustion chamber.

The convective heat transfer coefficient on the gas side h_g is computed separately for each station using the Bartz equation:

$$h_g(x) = \frac{0.026}{D_t^{0.2}} \left(\frac{D_t}{R_c} \right)^{0.1} \frac{\mu_0^{0.2} c_p}{Pr^{0.6}} \left(\frac{p_0}{c^*} \right)^{0.8} \left(\frac{A_t}{A_x} \right)^{0.9} \left(\frac{\rho_{ref}}{\rho_{e,x}} \right)^{0.8} \left(\frac{\mu_{ref}}{\mu_0} \right)^{0.8} \quad (38)$$

where the subscript 0 means the stagnation condition, e represents the free flow condition, and ref is a value obtained at reference temperature defined as the midpoint between the free flow temperature and the wall temperature. The curvature radius R_c was taken from the convergent side of the nozzle and used for the whole domain.

The heat transfer coefficient on the coolant side h_c is calculated using the Dittus-Boelter correlation⁷.

$$Nu = 0.023 Re^{0.8} Pr^{0.4} \implies h_c = \frac{Nu \cdot k_c}{D_{hyd}} \quad (39)$$

Considering the partial recovery of the total temperature due to boundary-layer deceleration effects, the adiabatic wall temperature can be characterized in terms of the recovery factor. The recovery factor is defined as the ratio between the actual adiabatic wall temperature rise and the ideal total temperature rise:

$$R_{rec} = \frac{T_w - T_\infty}{T_0 - T_\infty} \quad (40)$$

Considering a turbulent boundary layer, an empirical correlation is used:

$$R_{rec}(x) = \frac{1 + Pr^{1/3} \left(\frac{\gamma-1}{2} M_x^2 \right)}{1 + \frac{\gamma-1}{2} M_x^2} \quad (41)$$

The effective driving temperature is evaluated using the recovery temperature:

$$T_{aw}(x) = R_{rec}(x) T_0 \quad (42)$$

Heat transfer is modeled using a one-dimensional radial thermal resistance network:

$$R_{tot}(x) = \frac{1}{h_g(x)} + \frac{t_{tbc}}{k_{tbc}} + \frac{t_w}{k_{wall}} + \frac{1}{h_c(x)} \quad (43)$$

The local heat flux is then computed as:

$$q''(x) = \frac{T_{aw}(x) - T_c(x)}{R_{tot}(x)} \quad (44)$$

The temperature distribution across the wall is obtained for every section from:

$$T_{tbc} = T_{aw} - q'' \frac{1}{h_g} \quad (45)$$

⁷Considering that k_c is the thermal conductivity of water at the average temperature and assuming $D_{hyd} = 2$ mm

$$T_{wall,h} = T_{tbc} - q'' \frac{t_{tbc}}{k_{tbc}} \quad (46)$$

$$T_{wall,c} = T_{wall,h} - q'' \frac{t_w}{k_{wall}} \quad (47)$$

And the coolant temperature in the next section is obtained as:

$$T_c(x + \Delta x) = T_c(x) + \frac{q''(x) \cdot P(x)}{\dot{m}_w c_p} \quad (48)$$

where $P(x)$ is a heat transfer surface area estimated for a single section.

4.2.3 Algorithm for obtaining solutions

The heat transfer problem is nonlinear due to the dependence of the gas-side heat transfer coefficient on wall temperature. Therefore, an iterative procedure is adopted at each station:

1. An initial guess for the gas-side wall temperature is assumed.
2. The gas-side heat transfer coefficient h_g is computed using the Bartz equation (Equation 38).
3. The total thermal resistance R_{tot} (Equation 43) and corresponding heat flux $\dot{q}$ (Equation 44) are evaluated.
4. The TBC surface temperature is updated based on the computed heat flux.
5. Steps 2-4 are repeated until convergence is reached, i.e.:

$$|T_{tbc}^{(k+1)} - T_{tbc}^{(k)}| < 10^{-3}$$

6. Once convergence is achieved, the coolant temperature is updated (Equation 48).

Then the procedure is repeated for the next axial section. Once the solution for the entire domain is obtained, the temperatures of the TBC and the wall are compared with the maximal operating conditions. If the conditions are not satisfied, the mass flow rate of the coolant is increased and the whole calculation process is repeated.

4.2.4 Design variables discussion

The satisfaction of thermal and operational constraints is achieved by adjusting key design variables, including the coolant operating pressure, coolant mass flow rate, and the coolant flow velocity.

The increase of the operating pressure raises the boiling point of the cooling water. To prevent boiling on the interface of the cold side wall and the cooling water, the maximum temperature of the cold side must be lower than the boiling point, hence the increase of operating pressure enables relaxation of thermal constraint. At the same time, to decrease the load acting on the nozzle wall, the pressure of the coolant should be similar to the pressure of the hot gas inside the nozzle. Preferably lower, to avoid a contracting load acting on the wall.

The increase of both the coolant mass flow rate and the flow velocity affect the heat transfer favorably. Higher mass flow rate increases the heat capacity of the coolant while higher flow velocity increases the heat transfer coefficient of the coolant, lowering the total heat resistance of the system. Nevertheless, the final values of these variables are required to be realistic and define a system that is possible to be manufactured (e.g. the total area of the cooling channels).

4.3 Bare wall case

In order to assess the necessity of a TBC, a preliminary analysis was performed considering a bare metal wall configuration, i.e. without any insulating layer between the hot gas and the structural material. The wall material was chosen to be Inconel 718 for manufacturing purposes. The wall thickness was set to $t_w = 1$ mm. Thinner wall improves the heat transfer efficiency while 1 mm still ensures the structural integrity of the nozzle. The same computational model described in the previous section is applied, with the thermal resistance reduced to:

$$R_{tot} = \frac{1}{h_g} + \frac{t_w}{k_{wall}} + \frac{1}{h_c} \quad (49)$$

The objective is to verify whether the thermal constraint on the hot-side wall temperature can be satisfied by adjusting the available design variables, namely coolant pressure, mass flow rate, and flow velocity.

To evaluate the effectiveness of these parameters, a parametric analysis was conducted by varying coolant pressure and mass flow rate. Two scenarios were considered with flow velocity subsequently $v_w = 10$ m/s and $v_w = 30$ m/s, which makes the analysis consider a favorable, but unrealistic scenarios as the total area of the cooling channels is impossibly low even for low mass flow rates. The resulting maximum hot wall temperature is reported in Fig. 11.

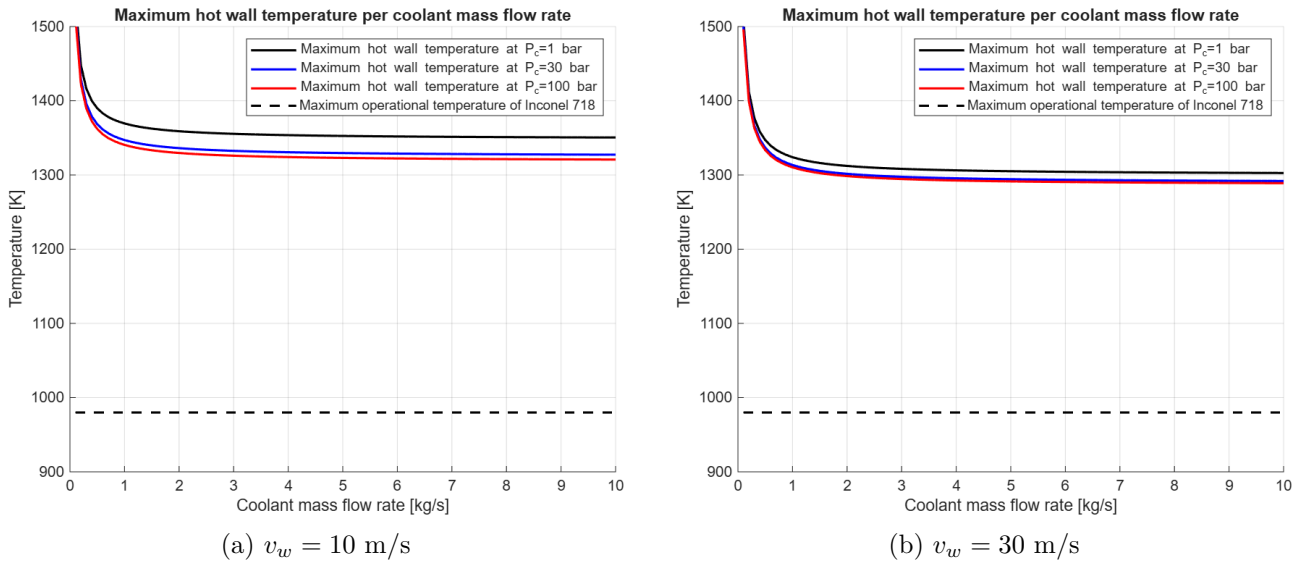


Figure 11: Bare case parametric analysis results

As can be seen, even in an unrealistically favorable scenario, the temperature of the hot side of the wall is significantly higher than the maximum operational temperature of the material. An asymptotic behaviour is present and further adjustment of the design variables won't satisfy the constraints, unless considering enormous and even more unrealistic values of the design variables. This means that the thermal conditions of the hot side of the wall are dominated by the hot gas flow in the nozzle and the coolant performance have a limited impact. Therefore, it can be concluded that a bare wall configuration is not sufficient to ensure safe operation within material temperature limits. This result motivates the introduction of a TBC, which increases the total thermal resistance and reduces the heat flux to the structural material.

4.4 TBC case

A similar analysis as in the previous section was conducted for a configuration with a $175 \mu\text{m}$ layer of YSZ with thermal conductivity $k = 1.55 \text{ W/m}\cdot\text{K}$ [3]. Only the scenario with a velocity of 5 m/s was sufficient to show a significant improvement with thermal conditions.

Figure 12a shows that the maximum temperature of the hot side of the wall was significantly decreased even though the velocity of the flow was reduced compared to the bare case. For higher operating pressures, the mass flow rate of the coolant of 1 kg/s is already enough to satisfy the safety condition. Further, an increase in the operating pressure widens the safety margin.

In addition to the hot-side safety conditions, the cold-side conditions must also be satisfied. As stated before, the cold-side temperature of the wall must be lower than the boiling point of the coolant. This condition is depicted in Figure 12b. While 1 bar of operating pressure is not enough to satisfy the requirement, for higher pressures, the safety condition is met with a considerable margin.

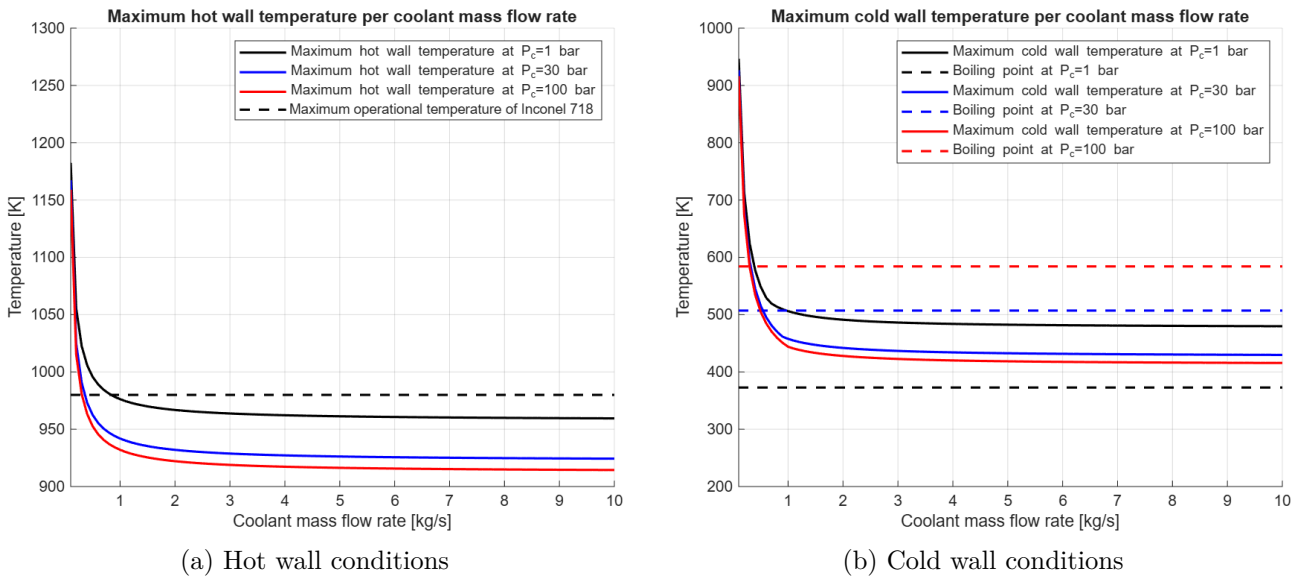


Figure 12: TBC case parametric analysis results

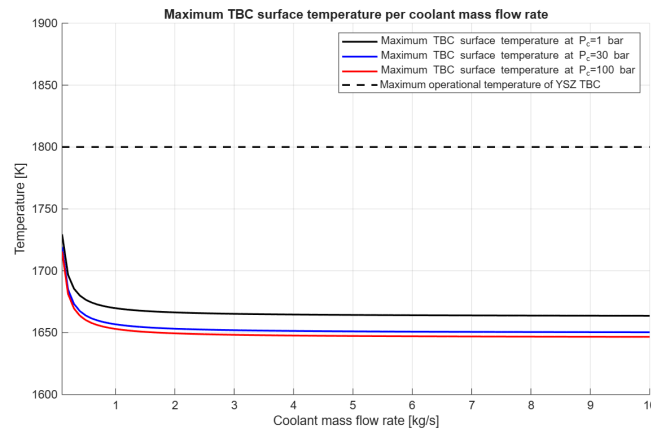


Figure 13: TBC surface thermal conditions

The surface temperature of TBC was also checked considering that the operational limit of YSZ is 1800 K . The temperature analysis is shown in Fig. 13. For all the operating points covered, the safety condition is met with a significant safety margin.

In conclusion, the application of TBC layer ensures the safety thermal conditions are met for a wide range of values of the design parameters. The final design parameters can now be chosen to satisfy the rest of the requirements as done in the next section.

4.5 Final design

The final cooling system configuration is evaluated through the temperature distribution (Figure 14) and heat flux (Figure 15) along the nozzle axis and the associated thermal and hydraulic performance parameters (Table 7). The results correspond to the selected operating conditions that satisfy all the design requirements.

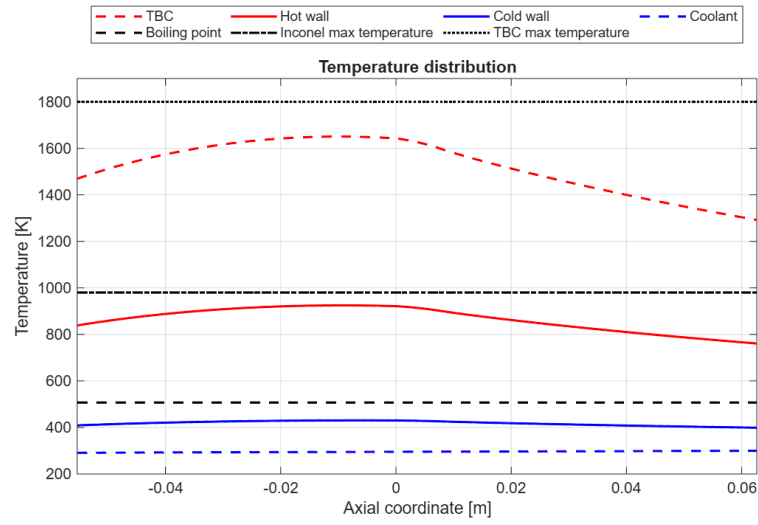


Figure 14: Temperature distribution

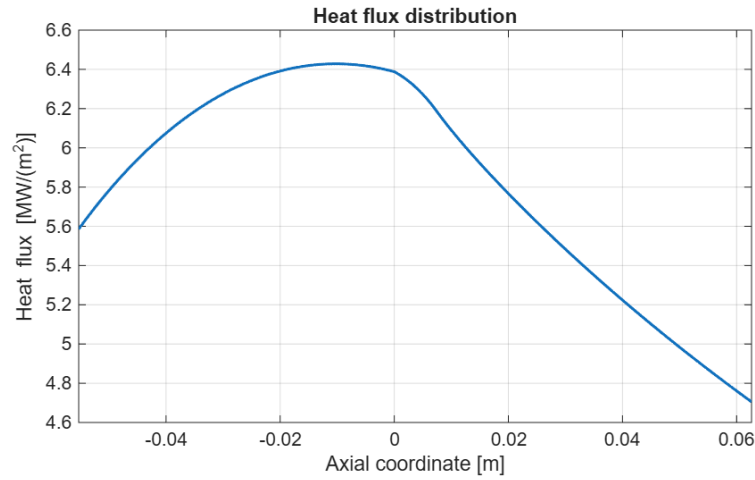


Figure 15: Heat flux distribution

Flow & Thermodynamic Parameters		Thermal Performance	
Inlet pressure P_{in} [bar]	30.13	Total heat load $\dot{Q}$ [kW]	270.66
Outlet pressure P_{out} [bar]	30.00	Max heat flux q''_{max} [MW/m ²]	6.43
Inlet temperature T_{in} [K]	291.15	Min heat flux q''_{min} [MW/m ²]	4.71
Outlet temperature T_{out} [K]	299.76	Approx. channel area A [m ²]	0.0016
Mass flow rate $\dot{m}$ [kg/s]	8.00	Max TBC wall temp [K]	1650.53
Coolant velocity v [m/s]	5.0	Max hot wall temp [K]	924.78
Reynolds number Re [-]	45240.89	Min hot wall temp [K]	760.89
Boiling point T_{boil} [K]	507.01	Max cold wall temp [K]	430.41
Required water mass m_w [kg]	232	Min cold wall temp [K]	398.95

Table 7: Cooling analysis results.

Overall, the designed cooling system satisfies all thermal and operational requirements, ensuring safe operation of the nozzle while enabling accurate estimation of the heat load through the measured coolant temperature rise.

Radiative heat transfer

The average effect of radiative heat flux was computed after the convective and conductive contributions to assess its relative importance in the overall thermal balance by using Equation 50⁸:

$$\bar{q}''_{rad} = \varepsilon \sigma \cdot (\bar{T}_{aw}^4 - \bar{T}_{tbc}^4) = 0.514 \text{ MW/m}^2 \quad (50)$$

This shows that the radiative heat flux remains significantly lower than the convective–conductive heat flux, contributing only a small fraction of the total heat transfer ($\approx 10\%$). Under the present operating conditions, heat transfer is strongly dominated by convection and conduction mechanisms. Therefore, the assumption of neglecting radiation for first-order design and sizing calculations can be reasonably justified, although radiative heat transfer remains relevant for higher-fidelity analyses or at elevated surface temperatures where its contribution may increase.

4.6 Design nozzle thermal protections

To protect the nozzle in the regions not covered by the cooling jacket, a carbon–phenolic ablative material was selected based on the local temperature levels, exposure time, and heat flux acting on these areas. The thickness sizing is based on Figure 16, which reports the time evolution of temperature across different thicknesses for a sample exposed to a heat flux of 6 MW/m² for 30 s, a condition that is similar to the one expected near the designed nozzle throat.

Considering carbon–phenolic with a thickness of 1 cm, the temperature at the Inconel 718 hot surface would be around 800 K, which is well within its structural integrity range [21] and significantly below its maximum short-term oxidation limit of 1253 K [22].

⁸considering YSZ emissivity $\varepsilon = 0.65$ and the average T_{aw} , T_{tbc} obtained from the aforementioned computation

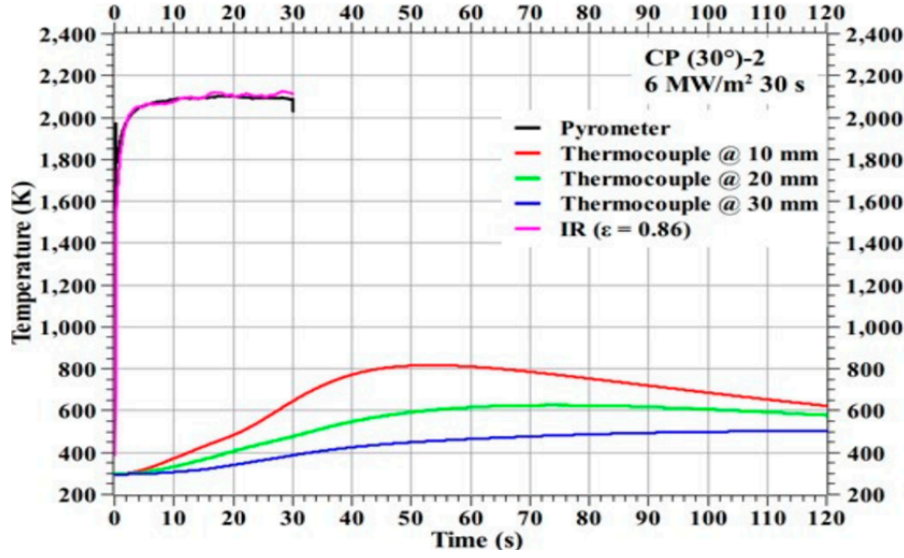


Figure 16: Temperature profile under carbon-phenolic protection [23]

The final thickness can be computed considering a regression rate of $r_r = 6.14 \text{ mm/min}$ [23] :

$$t_f = 1 \text{ cm} - r_r \cdot t_b = 7.44 \text{ mm} \quad (51)$$

The selected value for the carbon-phenolic thickness is a conservative choice, as the heat flux is lower in regions away from the nozzle throat, and potentially a smaller thickness may also satisfy the requirements, but since 1 cm is the smallest available thickness for which temperature measurements are reported, this value has been selected as the minimum allowable value.

4.7 Nozzle structural shell

The simplified membrane theory, already introduced in subsection 3.3, can also be applied to the nozzle geometry due to the thin-walled nature of the shell.

$$t_{noz} = \frac{P(x)R_{noz}(x)\eta}{\sigma_u} \quad (52)$$

Charts from [21] were used to evaluate the ultimate tensile strength of the Inconel 718 at different temperatures. In particular, the worst-case scenario curve was considered, corresponding to a temperature of 973.15 K (700 °C) and characterized by an approximate strength of $\sigma_u \approx 900 \text{ MPa}$, to which a safety factor of $\eta = 1.5$ is applied.

The required structural thickness was then computed by accounting for the local pressure and radius variation along the nozzle. As a preliminary design approximation, a constant thickness for the thermal protection layer was assumed along the entire nozzle axis⁹.

$$R_{noz} = R_{geo} + 1 \text{ cm} \quad (53)$$

The following graph shows the required minimum thickness along the nozzle axis:

⁹where R_{geo} is the geometrical radius of the nozzle computed in subsection 3.4

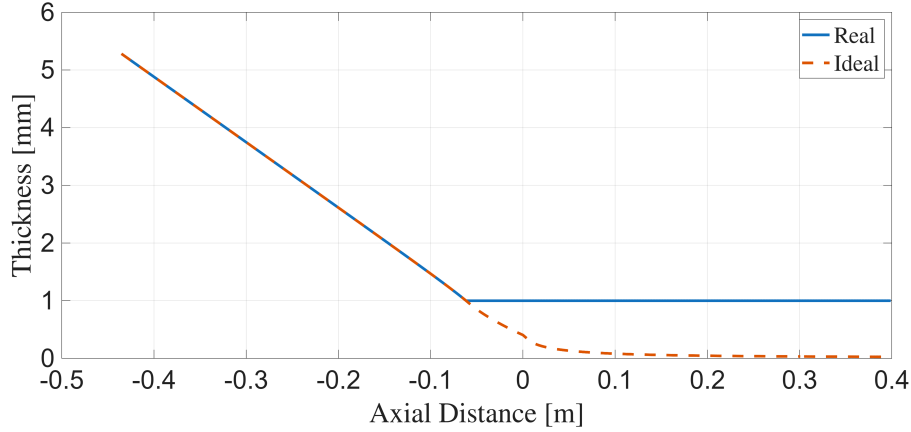


Figure 17: Structural shell thickness distribution (In718)

As illustrated in Figure 17, computed thicknesses below 1 mm were set to a minimum constant value of 1 mm due to technological manufacturing constraints and structural stiffness requirements. Ultimately it's important to notice that in the section of the nozzle covered by the cooling jacket the thickness of the wall is constant and equal to 1 mm, justifying the calculations done in subsection 4.1.

4.8 Technological design

The pressure drop in the cooling channels is evaluated using the Darcy–Weisbach formulation:

$$\Delta P = f \frac{L}{D_h} \frac{\rho V^2}{2} \quad (54)$$

where f is the Darcy friction factor, L is the channel length, D_h is the hydraulic diameter, ρ is the fluid density, and V is the mean flow velocity.

The friction factor for turbulent flow is obtained from the implicit Colebrook–White equation [24]:

$$\frac{1}{\sqrt{f}} = -2 \log_{10} \left(\frac{\varepsilon}{3.7 D_h} + \frac{2.51}{Re \sqrt{f}} \right) \quad (55)$$

where ε is the channel roughness and Re is the Reynolds number. The equation is solved iteratively to obtain f , which is subsequently used in the Darcy–Weisbach relation to compute the pressure drop. Value of roughness $\varepsilon = 35 \mu m$ is corresponding to an additively manufactured metal channel [25]. The channel size was estimated to have a square cross-section with 5 mm side length [26].

4.9 Design of the external supporting apparatus

The feeding system architecture is designed to provide a steady and controlled mass flow of cooling water to the engine's jacket. Given the target mass flow rate of 8 kg/s and the necessity to maintain a constant inlet pressure of 30.13 bar to ensure stable cooling performance, a pressure-fed configuration was selected.

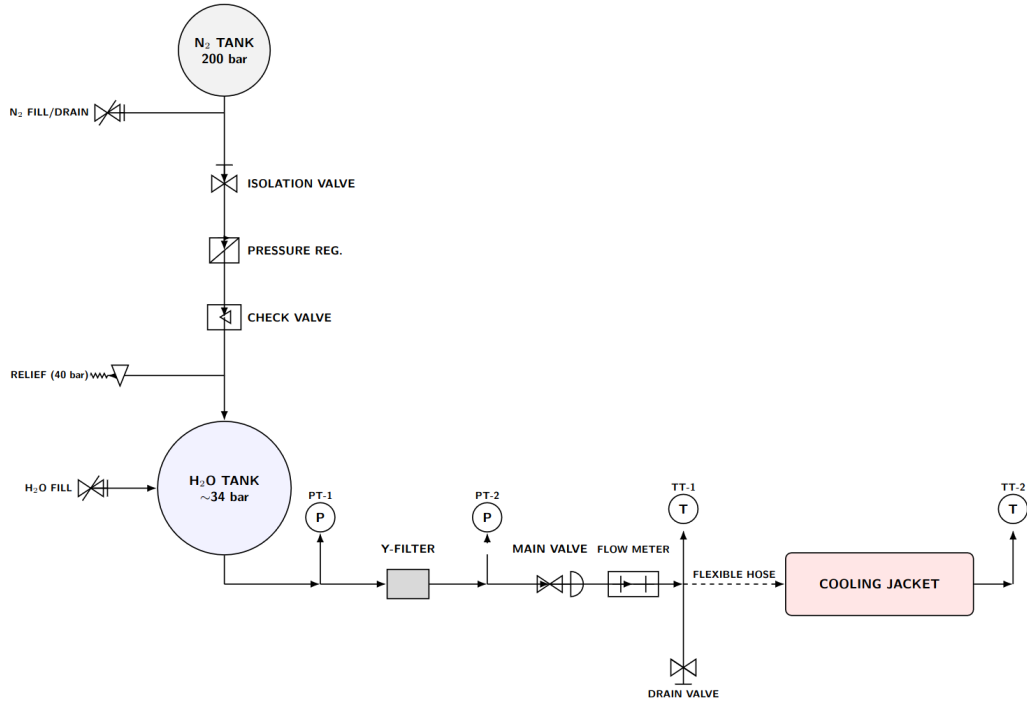


Figure 18: Supporting apparatus architecture

Line Sizing and Velocity Constraint: To meet the design prerequisite of a nominal flow velocity $v = 5$ m/s, which is consistent with the cooling jacket's internal fluid dynamics, the delivery line diameter was strictly sized. For a mass flow rate $\dot{m} = 8$ kg/s of water at 18°C, the theoretical internal diameter required is:

$$D_{int} = \sqrt{\frac{4 \cdot \dot{m}}{\pi \cdot \rho \cdot v}} \cdot 10^3 = 45.16 \text{ mm} \quad (56)$$

Given this value, the 1.5-inch Nominal Pipe Size (NPS) was selected, providing an actual internal diameter of approximately 45.5 mm and a resulting velocity of 4.92 m/s. This negligible deviation ($< 2\%$) ensures that the fully turbulent flow regime and heat transfer efficiency are maintained.

Notably, while the delivery line is sized at 1.5 inches to enforce the velocity constraint, the tank's primary outlet features a larger diameter. This design choice is intended to minimize entrance losses (often termed suction losses); the flow is then gradually accelerated to the target velocity via a conical reducer before entering the main 1.5-inch delivery line.

Liquid Feed and Control The liquid line is designed to handle the coolant flow while protecting the engine's structural integrity. The selection of the components is driven by the following functional requirements:

- **Filtration:** A Parker Conflow 36H Y-strainer filter [27], sized at 1.5 inches to match the line diameter, is positioned upstream of the main valve. It intercepts debris larger than 100 mesh, preventing the clogging of the narrow cooling channels and ensuring the stability of the pressure drop at the design velocity.
- **Flow Control:** A pneumatically actuated 1.5-inch Swagelok 68 series ball valve [28] manages the test initiation. The pneumatic actuation allows for a fast and repeatable opening transient, which is critical for reaching nominal cooling conditions before peak heat fluxes occur.

- **Instrumentation:** Two pressure transducers (PT-1 and PT-2) are strategically placed across the filter. This configuration allows for real-time monitoring of the pressure drop, serving as a health-monitoring system for the cooling circuit and providing data for the validation of the hydraulic model.

Structural Layout: As illustrated in Figure 18, the water tank features a bottom-outlet configuration. This design ensures a continuous liquid column and prevents gas ingestion until the required 232.29 kg of water are fully drained. The connection to the cooling jacket is finalized through a high-pressure flexible hose to decouple the test stand from structural vibrations.

Pressure Drop Analysis and System Pressurization: The operational pressures of the storage tanks were determined through a bottom-up hydraulic analysis, starting from the target requirement of 30.13 bar at the cooling jacket inlet. The total pressure to be supplied by the H_2O tank (P_{tank}) is calculated as:

$$P_{tank} = P_{jacket} + \sum \Delta P_{localized} + \Delta P_{distributed} + P_{margin} \quad (57)$$

where $\sum \Delta P_{localized}$ accounts for the losses across the 1.5-inch Parker filter and the Swagelok main valve, while $\Delta P_{distributed}$ represents the friction losses along the piping at the design velocity. As detailed in Table 8, the minimum required tank pressure is approximately 32.2 bar.

To ensure stable performance and compensate for potential filter fouling, the water tank operational pressure is set to 34 bar. This provides a sufficient buffer, allowing the pressure regulator to maintain a constant delivery pressure despite the increased hydraulic resistance.

Component	Model/Series	Size	Key Parameter	ΔP [bar]
Main Valve	Swagelok SS-67TF24 [28]	1.5"	$C_v = 100$	0.12
Pressure Regulator	Tescom 26-1200 [29]	1/2"	$P_{in,max} = 400$ bar	-
Y-Strainer Filter	Parker Conflow 36H [27]	1.5"	From Perf. Curve	1.10
Check Valve	Swagelok SS-CHS8-1/3 [30]	1/2"	$P_{cracking} = 0.03$ bar	0.03
Isolation Valve (N_2)	Swagelok SS-63TS8 [28]	1/2"	$C_v = 7.5$	negl.
Fill & Drain Valves	Swagelok 40G Series [31]	1/2"	Manual Service	-
Safety Relief (H_2O / Low Pres.)	Swagelok R3A Series [32]	1/4"	Set @ 38 bar	-
Piping & Fittings	AISI 316L [33]	1.5"	$v \approx 5$ m/s	0.80
Total System Loss				2.05

Table 8: Summary of selected feeding system components and hydraulic parameters.

Detailed Piping Loss Estimation To account for the geometric complexity of the test stand—including bends, T-junctions, and the high-pressure flexible hose—an equivalent length $L_{eq} = 6$ m was conservatively assumed. The distributed pressure drop along the AISI 316L stainless steel network [33] is computed using the Darcy-Weisbach relation:

$$\Delta P_{distributed} = f \cdot \frac{L_{eq}}{D_{int}} \cdot \frac{\rho \cdot v^2}{2} \cdot 10^{-5} \approx 0.80 \text{ bar} \quad (58)$$

where the actual flow velocity $v = 4.92$ m/s and the internal diameter $D_{int} = 45.5$ mm are applied. To maintain consistency with the cooling jacket analysis, the Darcy friction factor ($f \approx 0.018$) for the fully turbulent flow regime ($Re \approx 2.1 \times 10^5$, based on water dynamic viscosity $\mu = 1.053$ mPa · s at 18°C) is obtained from the implicit Colebrook-White equation [24]. Incorporating the certified surface roughness ($\epsilon = 0.015$ mm) characteristic of drawn seamless tubing, the equation is solved iteratively to obtain f , confirming the localized and distributed loss budget.

Gas Pressurization Sizing and Optimization Following the bottom-up hydraulic analysis, the pressurization system must be sized to meet the end-of-burn boundary conditions. First, the total internal volume of the water tank (V_{t-f}) is derived from the required liquid mass (232 kg) and the water density at 18°C ($\rho \approx 998.6 \text{ kg/m}^3$), incorporating a 10% ullage volume to prevent gas ingestion and allow for thermal expansion:

$$V_{t-f} = \frac{m_{H2O}}{\rho_{H2O}} \cdot 1.1 \approx 0.255 \text{ m}^3 \quad (59)$$

When the water tank is fully depleted at the end of the burn, the residual pressure in the Nitrogen (N_2) vessel (P_{PG-f}) must still overcome the water tank setpoint (34 bar) plus the cumulative gas line losses. These include the check valve cracking pressure ($\Delta P_{check} = 0.03 \text{ bar}$) and the regulator's operational droop ($\Delta P_{reg} \approx 3 \text{ bar}$). This latter value is estimated from the manufacturer's flow curves [29], anticipating the pressure decay caused by the high volumetric gas flow rate required to displace the water at 8 kg/s. This balance establishes a minimum required final gas pressure of $P_{PG-f} \approx 37 \text{ bar}$. To determine the initial volume of the pressurant gas tank (V_{PG-i}), an initial storage pressure (P_{PG-i}) must first be selected. Since the required volume is inversely dependent on the selected initial pressure, a sensitivity analysis was performed to evaluate this trade-off. As shown in Figure 19, the required volume follows a hyperbolic decay as the initial pressure increases. Beyond 200 bar, the curve flattens significantly into a plateau; further increases in pressure yield marginal volume reductions while exponentially increasing the structural weight and the risk of gas leakage. Therefore, $P_{PG-i} = 200 \text{ bar}$ represents the optimal design point, providing the best compromise between gas storage density, system footprint, and the use of standard commercial hardware.

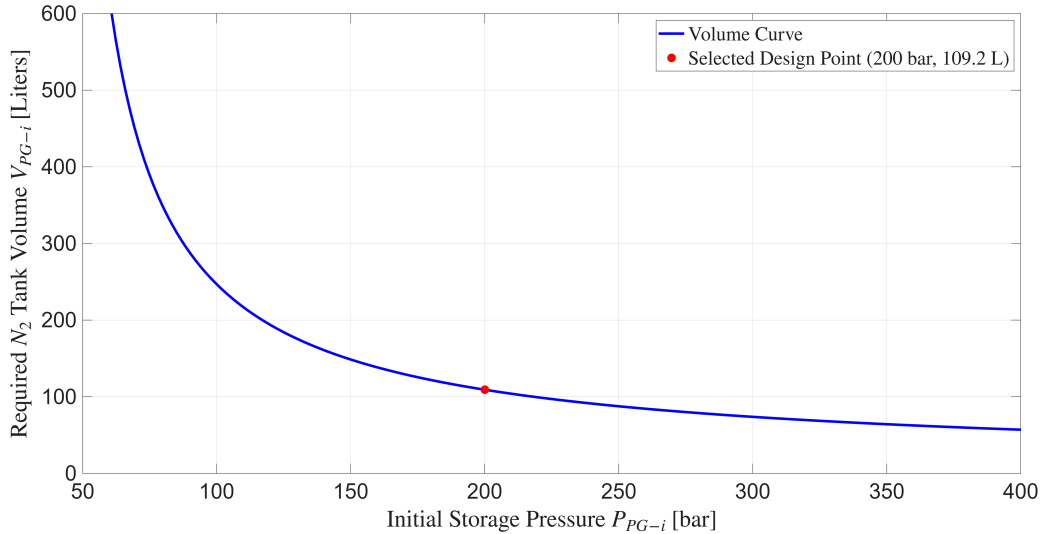


Figure 19: Nitrogen storage volume as a function of the initial pressure

Having established the optimal initial pressure, the general mass conservation approach is applied to compute the exact vessel volume. During the rapid 29-second blowdown from the 200 bar state, the expansion of Nitrogen causes a significant temperature drop. Assuming an isentropic expansion ($\gamma = 1.4$), the final gas temperature decreases from ambient conditions ($T_{PG-i} = 293.15 \text{ K}$) to approximately $T_{PG-f} = 181 \text{ K}$.

Equating the initial mass of the pressurant to the final mass expanded across both the gas and water volumes, the final total volume occupied by the gas (V_{PG-f}) is derived as:

$$V_{PG-f} = \frac{V_{t-f}}{\left(1 - \frac{P_{PG-f}}{P_{PG-i}} \cdot \frac{T_{PG-i}}{T_{PG-f}}\right)} \quad (60)$$

where $V_{t-f} \approx 0.255 \text{ m}^3$ is the previously calculated total volume of the water tank. Solving this relation yields $V_{PG-f} \approx 0.364 \text{ m}^3$. The isolated internal volume required for the high-pressure Nitrogen source is then calculated by subtracting the water tank volume:

$$V_{PG-i} = V_{PG-f} - V_{t-f} = 0.109 \text{ m}^3 \quad (61)$$

This calculation definitively sizes the gas storage volume at approximately 109 L, ensuring the system can execute the test without dropping below the operational pressure floor.

Gas Management Hardware To physically manage the pressurization cascade from the aforementioned 200 bar Nitrogen source down to the 34 bar water tank, a dedicated regulation line is implemented. The pressure is stepped down via a high-performance Tescom 26-1200 regulator [29]. A Swagelok check valve [30] is installed immediately downstream to prevent water backflow into the pneumatic regulation line during transient operations. Finally, passive safety is guaranteed by a proportional relief valve [32] set at 38 bar ($> 10\%$ above the operational setpoint) to safely vent the gas and protect the structural integrity of the water tank in the event of a regulator failure.

Structural Tank Sizing and Mass Estimation: A preliminary structural sizing was conducted based on the ultimate tensile strength (F_{ut}) of AISI 316L stainless steel ($F_{ut} \approx 485 \text{ MPa}$). To ensure safe operation, a safety factor of 2 was applied to define the burst pressure ($P_b = 2 \cdot P_{oper}$) for both vessels.

For the water tank ($P_{oper} = 34 \text{ bar}$, $P_b = 68 \text{ bar}$), assuming a spherical geometry to minimize structural mass, the internal radius is $r_{t-H_2O} \approx 0.393 \text{ m}$. Applying thin-wall pressure vessel theory, the required minimum wall thickness t_{t-H_2O} is:

$$t_{t-H_2O} = \frac{P_b \cdot r_{t-H_2O}}{2 \cdot F_{ut}} \approx 2.7 \text{ mm} \quad (62)$$

This results in an estimated structural mass of $M_{t-H_2O} \approx 42 \text{ kg}$.

Applying the identical methodology to the high-pressure Nitrogen vessel ($P_{oper} = 200 \text{ bar}$, $P_b = 400 \text{ bar}$, $r_{t-PG} \approx 0.296 \text{ m}$), the resulting minimum wall thickness is $t_{t-PG} \approx 12.2 \text{ mm}$. The mass of the Nitrogen spherical tank is consequently estimated at $M_{t-PG} \approx 107 \text{ kg}$, confirming that the high-pressure gas storage constitutes the primary structural mass driver of the entire feeding apparatus.

Temperature and heat flux measurements: The heat load transferred to the cooling jacket is experimentally determined using a calorimetric approach based on the coolant temperature rise. Assuming single-phase flow and negligible heat losses to the surroundings, the total heat transfer rate is obtained from an energy balance on the coolant:

$$\dot{Q} = \dot{m} c_p (T_{out} - T_{in}) \quad (63)$$

where $\dot{m}$ is the coolant mass flow rate, c_p is the specific heat capacity of water, and T_{in} and T_{out} are the inlet and outlet temperatures of the cooling jacket, respectively.

Given the known heat transfer area A , the average heat flux can be estimated as:

$$\bar{q}'' = \frac{\dot{Q}}{A} \quad (64)$$

Temperature measurements are performed using fast-response thermocouples installed at the inlet and outlet sections of the cooling line. Given the relatively small temperature rise ($\Delta T \approx 9 \text{ K}$), high measurement accuracy is required. Therefore, sensors with an uncertainty below $\pm 0.1 \text{ K}$ are recommended to limit the error on the estimated heat load.

OMRON E52-CA thermocouples [34] were selected as they provide fast-response, high-precision direct immersion temperature measurements suitable for liquid coolant flow applications. The thermocouples are mounted using compression fittings on instrumentation bosses, allowing direct insertion of the sensing junction into the main flow. The probes are oriented such that the tip faces upstream into the flow to minimise flow separation effects and improve dynamic thermal response.

The use of thermowells was avoided due to their increased thermal inertia, which introduces significant response lag and damping of transient temperature variations.

The mass flow rate is measured using a calibrated flow meter (Proline Promass F 300 Coriolis flowmeter [35]). The combined uncertainty on the heat transfer rate is primarily driven by the temperature measurement, due to the small temperature difference.

5 Critiques and improvements

Possible improvements in the modeling and design of the SRM system, together with a discussion of the main limitations of the present analysis, as described in the following considerations:

- **Improvements in SRM burning modeling**

Erosive burning was not considered in the present analysis but could be relevant in a more detailed model. It may occur in regions of high mass flux, particularly near the last section of the grain, where flow velocities are higher, leading to locally increased regression rates and a consequent modification of the chamber pressure evolution. This phenomenon is described by J.M. Lenoir and G.Robillard [36] who propose a composite formula for the burning rate: $r = r_p + r_e$. Here, r_p represents the pressure dependent burning rate, while r_e accounts for the erosive component. This last term incorporates the influence of both pressure and gas velocity, a physical coupling - especially at the beginning of the firing when the port area is narrowest - that is not accounted for in the simplified model used in this report.

- **Improvements in thermo-structural analysis**

Thermal stresses in the Inconel nozzle wall, driven by high temperature gradients across a 1 mm thickness, were not included in the present model but may become critical. Accounting for them could require an increased structural thickness that could reduce heat flux and increase surface temperature, possibly rendering non effective the current design and requiring a dual-layer TBC configuration, due to this thermo-structural coupling effect.

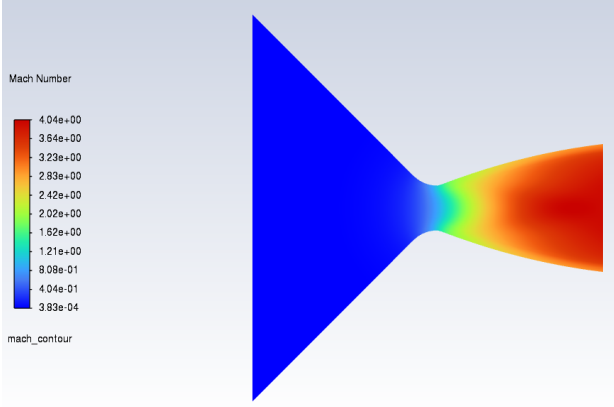
- **Improvements in nozzle ablative protection**

The non-uniform ablation of the nozzle protective material was not explicitly considered in the present analysis but could affect the design in a more detailed model. In particular, the ablative thickness could be tailored according to the varying heat flux levels, which are lower farther from the throat and higher in its vicinity. This could lead to a more optimized design, also reducing the nozzle shell diameter in regions that need less ablative protection.

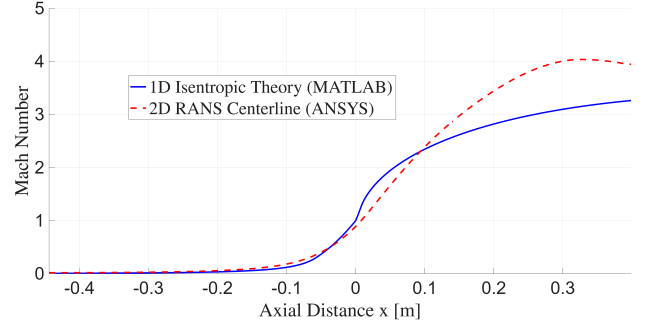
- **Viscous effects and 2D flow limitations**

The baseline nozzle design relies on a quasi-1D isentropic flow assumption, which assumes uniform fluid properties across any cross-section. However, by superimposing the analytical Mach number profile with the centerline results from the 2D axisymmetric Navier-Stokes simulation (ANSYS), a significant difference is observed (see Figure 20). The CFD centerline Mach number is noticeably higher than the 1D prediction. This discrepancy physically highlights the limitations of the quasi-1D model: in the real 2D viscous flow, the boundary layer development

significantly slows the gas near the nozzle walls. To satisfy mass conservation, the core flow along the centerline is forced to accelerate to higher velocities than the cross-sectional average. Furthermore, expansion waves originating from the throat curvature intersect at the symmetry axis, locally increasing the centerline Mach number. This confirms that while the 1D model is suitable for initial sizing, 2D CFD validations are strictly required to accurately capture the real flow field distribution.



(a) Mach number contour plot evaluated via 2D axisymmetric CFD.



(b) Centerline Mach number: analytical (1D) vs numerical (2D).

Figure 20: Comparison between isentropic (1D) theory and CFD simulation.

6 Conclusion

The developed solid rocket motor model provides consistent and realistic predictions of the system performance under steady-state assumptions. The ballistic characterization enabled accurate estimation of the burning rate, leading to reliable values of thrust, specific impulse, and total impulse that meet the target requirements. The uncertainty analysis shows that variations in ballistic parameters and operating conditions mainly affect the MEOP and burning time, but the system remains within acceptable margins, confirming the robustness of the design.

Thermal analysis highlights that the heat flux in the nozzle throat region is extremely high and cannot be sustained by a bare metallic wall, even when combined with an actively cooled water jacket. The introduction of a thermal barrier coating reduces the heat transferred to the structure, maintaining wall temperatures within safe limits. The cooling system is capable of removing the imposed thermal load while avoiding coolant boiling under the selected operating conditions. The external cooling apparatus, designed as a self-contained pressure-fed system, satisfies the requirement of independent operation throughout the full firing duration.

Overall, for a preliminary design, the results obtained demonstrate that the proposed configuration achieves a sufficient balance between performance, thermal protection, and operational feasibility, providing a reliable solution for high heat-flux rocket nozzle environments.

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Declaration of authorship

We, the undersigned, declare that this report titled "**Thrust Issues**" and the work presented in it are our own. We confirm that:

- This work was done wholly while in candidature for a Master's degree at Politecnico di Milano.
- Where we have consulted the published work of others, this is always clearly attributed.
- Where we have quoted from the work of others, the source is always given.

While each member's main technical contributions are outlined in the table below, the project was a highly collaborative effort, with all authors contributing to sections beyond their primary assigned tasks.

Declaration of authorship		
Jack	Challita	TBC literature review, Design of the cooling jacket
Bruno	Macchia	Design of nozzle, cooling jacket and supporting apparatus
Krzysztof	Mazurek	Design of supporting apparatus and chemical computations
Alessandro	Ponti	Design of composition, grain, performance and ablative protections
Bastien	Thomas	BATES literature review, Design of the grain & Combustion chamber
Tymon	Tomczykowski	Design of the cooling jacket and supporting apparatus

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